ITCS216 Project Phase I

Subtitle: Jett Fitness

Section: 1
Group: Group 7
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Team Members (Sorted by Student ID)

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1. Introduction: Business Domain Overview

Jetts Fitness is an Australia-based chain of fitness centers founded in Queensland in 2007, operating across countries including New Zealand, the Netherlands, the UK, India, and Thailand. The gym provides 24/7 access, no lock-in contract flexibility, experienced staff, and free monthly member classes. Facilities are categorized into Red (standard facilities) and Black (larger, luxury tier) gyms, accessible via electronic member cards. Jetts offers tiered memberships including Freedom (1,750 Baht/month), Flexi (1,650 Baht/month, 6-month minimum), and Stay Fit (1,550 Baht/month, 12-month minimum). The project scope encompasses core administrative operations across member onboarding, monthly billing, and transaction processing.

2. Workflow and Operational Activities

2.1 Membership Registration

1. Who initiates the action?
    a. Customer and the Front-Desk Staff are the action initiators.
2. What data is captured or updated?
    a. Customer's personal identity information, emergency contacts, payment method credentials, selected Membership plan, registration timestamp.
3. What state change occurs in the system?
    a. User/Member State: Customer -> Member
    b. Access Control State: No access -> Active Access.

2.2 Monthly Invoice

1. Who initiates the action?
    a. Billing System, the automated system executing at the start of a billing cycle.
2. What data is captured or updated?
    a. Invoice number, issue date, billing cycle period, line items, total amount due, and payment due date.
    b. The member's account balance ledger is increased by the invoice amount, and the next billing timestamp is updated.
3. What state change occurs in the system?
    a. Invoice Stat: Null -> Issued -> Unpaid / Due
    b. Account Standing: Settled -> Outstanding Balance Pending.

2.3 Payment & Receipt

1. Who initiates the action?
    a. Billing Engine (processes the transactions)
    b. Customer (If paying manually)
    c. Front-Desk Staff (If processing cash, POS machine, and bank transfer)
2. What data is captured or updated?
    a. Payment transaction ID, payment gateway reference code, payment method used, payment timestamp, payment status, and receipt ID.
    b. Member Account balance.
3. What state change occurs in the system?
    a. Invoice State: Unpaid / Due -> Paid / Settled
    b. Receipt State: Null -> Generated / Issued
    c. Account Standing: Payment Due -> Active Renewed

3. Business Rules and Constraints

Process:	Membership Registration
Business rules:	1. A customer can have one active membership contract only. (1:1) 2. A membership plan can be selected by many customers. (1:M) 3. A member can access multiple gyms that are the same tier. (1:M) 4. A club can register many members. (1:M) 5. A customer needs to provide health certification to become a member. (1:1) 6. A member can have multiple emergency contacts. (1:M) 7. A payment method can settle many subscriptions. (1:M) 8. A member record must contain a unique member ID, first name, last name, ID number, address, phone number, email address, and timestamp. 9. A customer must be at least 16 years old to register and become a member. 10. A member must choose between “No Lock-in Flexible” and “Fixed-term commitment”.

Business Process Name: Member Registration		
Rule No. 1	**Rule Statement**	**Relationship**
Forward/Active	A customer can hold only one active membership contract.	1:1
Backward/Passive	An active membership contract can be held by only one customer.	1:1
Conclusion	A customer can hold only one active membership contract	1:1
Rule No. 2	**Rule Statement**	**Relationship**
Forward/Active	A customer can enroll in only one membership plan at a time.	1:1
Backward/Passive	A membership plan can be selected by many customers.	1:M
Conclusion	A membership plan can be selected by many customers.	1:M
Rule No. 3	**Rule Statement**	**Relationship**
Forward/Active	A member can access multiple gyms that are the same tier.	1:M

Backward/Passive	A gym within a tier can be accessed by many members.	1:M
Conclusion	Each member can access many gyms of the same tier.	M:N
Rule No. 4	**Rule Statement**	**Relationship**
Forward/Active	A club can register many members.	1:M
Backward/Passive	A member can be registered at only one club.	1:1
Conclusion	A club can register many members.	1:M
Rule No. 5	**Rule Statement**	**Relationship**
Forward/Active	A customer must provide only one health certificate.	1:1
Backward/Passive	A health certificate is provided by only one customer.	1:1
Conclusion	A customer must provide only one health certificate.	1:1
Rule No. 6	**Rule Statement**	**Relationship**
Forward/Active	A member can have multiple emergency contacts.	1:M
Backward/Passive	An emergency contact is associated with only one member.	1:1
Conclusion	A member can have multiple emergency contacts.	1:M
Rule No. 7	**Rule Statement**	**Relationship**
Forward/Active	A payment method can settle many subscriptions.	1:M
Backward/Passive	A subscription can be settled by only one payment method	1:1
Conclusion	A payment method can settle many subscriptions.	1:M
Rule No. 8		
A member record must contain a unique member ID, first name, last name, ID number, address, phone number, email address, and timestamp.		
Rule No. 9		
A customer must be at least 16 years old to register and become a member.		
Rule No. 10		
A member must choose between “No Lock-in Flexible” and “Fixed-term commitment”.		

Process:	Monthly Invoice
Business rules:	1. A member can receive many monthly invoices. (1:M) 2. A monthly invoice can contain many line items. (1:M) 3. A monthly invoice must be linked to an active membership. (1: M) 4. Add-on services (personal trainer) can be added to many monthly invoices. (M:N) 5. A club branch can issue many monthly invoices. (1:1) 6. A tax rate policy can apply to many monthly invoices. (1:M) 7. Monthly invoices can apply many discount vouchers. (M:N) 8. A monthly invoice must have a unique Invoice_ID, Service_ID, issue date, payment due date, billing cycle, subtotal, tax rate, and total payable amount. 9. An invoice item must include Item_ID, Invoice_ID, service SKU, description, price, quantity, discount, and total price of the item. 10. If an invoice payment is overdue, the invoice status should be changed to Past_Due, and the electronic card should be frozen.

Business Process Name: Monthly Invoice		
Rule No. 1	**Rule Statement**	**Relationship**
Forward/Active	A member can receive many monthly invoices.	1:M
Backward/Passive	A monthly invoice can be issued to only one member.	1:1
Conclusion	A member can receive many monthly invoices.	1:M
Rule No. 2	**Rule Statement**	**Relationship**
Forward/Active	A monthly invoice can contain many line items.	1:M
Backward/Passive	A line item can belong to only one monthly invoice.	1:1
Conclusion	A monthly invoice can contain many line items.	1:M
Rule No. 3	**Rule Statement**	**Relationship**
Forward/Active	A monthly invoice must be linked to an active membership.	1:1
Backward/Passive	An active membership can be linked to many monthly invoices.	1:M
Conclusion	An active membership can be linked to many monthly invoices.	1:M

Rule No. 4	**Rule Statement**	**Relationship**
Forward/Active	Add-on services can be added to many monthly invoices.	1:M
Backward/Passive	A monthly invoice can include many add-on services.	1:M
Conclusion	Each add-on service may appear on many invoices, and each invoice may include many add-on services.	M:N
Rule No. 5	**Rule Statement**	**Relationship**
Forward/Active	A club branch can issue many monthly invoices.	1:M
Backward/Passive	A monthly invoice can be issued by only one club branch.	1:1
Conclusion	A club branch can issue many monthly invoices.	1:M
Rule No. 6	**Rule Statement**	**Relationship**
Forward/Active	A tax rate policy can apply to many monthly invoices.	1:M
Backward/Passive	A monthly invoice can apply only one tax rate policy.	1:1
Conclusion	A tax rate policy can apply to many monthly invoices.	1:M
Rule No. 7	**Rule Statement**	**Relationship**
Forward/Active	A discount voucher can be applied to many monthly invoices	1:M
Backward/Passive	A monthly invoice can apply many discount vouchers.	1:M
Conclusion	Many invoices apply many discount vouchers.	M:N
Rule No. 8		
A monthly invoice must have a unique Invoice_ID, Service_ID, issue date, payment due date, billing cycle, subtotal, tax rate, and total payable amount.		
Rule No. 9		
An invoice item must include Item_ID, Invoice_ID, service SKU, description, price, quantity, discount, and total price of the item.		
Rule No. 10		
If an invoice payment is overdue, the invoice status should be changed to Past_Due, and the electronic card should be frozen.		

Process:	Payment & Receipt
Business rules:	1. An invoice can be paid by multiple split transactions. (1:M) 2. A refund must correspond to the original transaction. (1:1) 3. A front-desk staff member can record multiple counter in-person payments. (1:M) 4. A customer can receive many receipts. (1:M) 5. One receipt can only be referred to by one payment(1:1) 6. A payment must contain Payment_ID, Invoice_ID, payment timestamp, payment method type, total amount paid, and payment status. 7. Completed payments and issued receipts cannot be deleted or modified. 8. The system automatically charges the member's default card on the invoice date. 9. If the automatic charge fails, it should retry charging up to 3 times. 10. A receipt must contain a Receipt_ID, Payment_ID, issue_date, member name, business name, tax_ID, amount paid, and part of the card number.

Business Process Name: Payment & Receipt		
Rule No. 1	**Rule Statement**	**Relationship**
Forward/Active	An invoice can be paid by multiple split transactions.	1:M
Backward/Passive	A transaction belongs to only one invoice.	1:1
Conclusion	An invoice can be paid by multiple split transactions.	1:M
Rule No. 2	**Rule Statement**	**Relationship**
Forward/Active	A refund must correspond to the original transaction.	1:1
Backward/Passive	An original transaction can have at most one refund.	1:1
Conclusion	Each refund is linked to exactly one original transaction, and each transaction has at most one refund.	1:1
Rule No. 3	**Rule Statement**	**Relationship**
Forward/Active	A front-desk staff member can record multiple counter in-person payments.	1:M
Backward/Passive	A counter in-person payment is recorded only by one staff member.	1:1

Conclusion	A front-desk staff member can record multiple counter in-person payments.	1:M
Rule No. 4	**Rule Statement**	**Relationship**
Forward/Active	A customer can receive many receipts.	1:M
Backward/Passive	A receipt is issued to only one customer.	1:1
Conclusion	A customer can receive many receipts.	1:M
Rule No. 5	**Rule Statement**	**Relationship**
Forward/Active	One receipt can only be referred to by one payment.	1:1
Backward/Passive	One payment can only generate one receipt.	1:1
Conclusion	Each receipt corresponds to exactly one payment, and each payment produces exactly one receipt.	1:1
Rule No. 6		
Completed payments and issued receipts must remain unchanged and cannot be deleted or modified.		
Rule No. 7		
The system automatically charges the member’s default card on the invoice date.		
Rule No. 8		
If the automatic charge fails, it should retry charging up to 3 times.		
Rule No. 9		
A receipt must contain a Receipt_ID, Payment_ID, issue_date, member name, business name, tax_ID, amount paid, and part of the card number.		
Rule No. 10		
A payment record must contain Payment_ID, Invoice_ID, payment timestamp, payment method type, total amount paid, and payment status.		

4. References

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